# Sankalp Parashar

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## Education

2020–2024 **B.Tech. (Honours) in Computer Science & Engineering**, IIT Bombay CPI: 9.89/10
Relevant Courses: Automatic Speech Recognition, AI and Machine Learning, Reinforcement Learning, Advanced Image Processing, Data Structures and Algorithms

## Publications

[1] **Sankalp Parashar**, Preethi Jyothi, Sunita Sarawagi, "Ask, Don't Guess: Targeted Clarification for Speech Recognition Errors," *Under review at NAACL 2027.* arXiv:2610.24809

## Research Experience

**Targeted Clarification for Speech Recognition** June 2026 – Present
*Prof. Sunita Sarawagi and Prof. Preethi Jyothi, IIT Bombay*

- Developed PARDON, a framework that reduces errors in streaming ASR by flagging acoustically degraded spans, asking the speaker to repeat them and using the repetition to correct the transcript
- Devised a new supervision technique that uses clean audio to estimate how the transcript would change on re-utterance, so that the detectors flag only the errors a repetition is likely to fix
- Identified up to **52% more erroneous words** than the strongest baseline at high precision, and achieved up to a **24% relative WER reduction** under a fixed clarification budget on far-field speech

**ASR for Regionally Accented Hindi Medical Speech** September 2026 – Present
*Prof. Srikanth Raj Chetupalli, IIT Bombay*

- Improving ASR on regionally accented and dialectal Hindi conversations between community health workers and patients in rural primary healthcare, as part of the DISPLACE-M challenge
- Adapted diarization conditioning to IndicConformer to transcribe longer multi-speaker segments with cross-turn context, achieving a **15% relative CER reduction** with oracle diarization

**Language Identification for Code-Switched ASR** July 2023 – April 2024
*B.Tech. Thesis, Prof. Preethi Jyothi, IIT Bombay*

- Analyzed failure modes of state-of-the-art multilingual ASR on code-switched speech, identifying language identification as a key weakness, and raised top-two accuracy from 16% to **81%** using an ECAPA-TDNN
- Devised a method to condition an RNN-T recognizer on the predicted distribution over languages, extending its default conditioning on a single language to handle code-switched speech

**Variations of the Voronoi Game and Art Gallery Problem** Summer 2022
*Research Internship, Prof. Sándor Fekete, TU Braunschweig*

- Conducted a literature review of variations of the Voronoi Game and the Art Gallery Problem
- Devised a winning second-player strategy for a two-move Voronoi Game with linearly fading influence

## Honors & Awards

- **Rank 7** in the **Department of Computer Science**, IIT Bombay, in a batch of 160+
- **All India Rank 29** in **JEE Advanced 2020** among 0.24 million eligible candidates
- **All India Rank 74** in **JEE Main 2020** among 1.17 million candidates
- Awarded gold medal in **Indian National Chemistry Olympiad**, given to top 46 candidates (2020)
- **AP grade** (top 1%) in 6 courses incl. **Game Theory**; **sole AP of 160** in Programming Paradigms Lab
- **IEEE OES Young Researchers' Prize** (2021) as part of the AUV-IITB team

## Work Experience

**Quantitative Strategist, Quadeye Securities** June 2024 – June 2026

- Solely drove the ideation, development and deployment of market-making strategies for the team in Indian F&O, building the generalized framework underlying all of them
- Automated daily generation of reports that monitored likely causes of mismatch between simulation and production, identifying and solving major causes, including cutting tick-to-trade latency from **3$\mu$s to 1$\mu$s**
- Supervised the scaling of the framework across multiple asset classes and new markets, with the strategies achieving a combined **annualized Sharpe ratio of 12**

**Quantitative Strategy Intern, Quadeye Securities** Summer 2023

- Implemented a generalized variant of regression trees in Python to combine market signals, trained on a loss that balances prediction error against a complexity penalty to keep the model interpretable
- Raised the correlation of predictions with market returns to **15%**, up from 6% with linear regression

## Selected Projects

**Matsya, Autonomous Underwater Vehicle (AUV-IITB)** January 2021 – September 2022
*Senior Software Developer, Prof. Leena Vachhani and Prof. Hemendra Arya, IIT Bombay*

- Improved detection mAP of the vehicle's vision module by upgrading it from YOLOv3 to YOLOv4, integrating the new model into the legacy vision pipeline
- Reduced travel time between setpoints by 40% by optimizing the waypoint navigation algorithm, and implemented LP-based thruster force allocation in C++ using GLPK
- Secured 2nd place in video presentation at RoboSub 2021, among 50+ teams from 12 countries

**Wildcard CTC for Partially Labeled Speech** Spring 2023
*Course Project, Automatic Speech Recognition, Prof. Preethi Jyothi, IIT Bombay*

- Reimplemented Wildcard CTC, which trains ASR on transcripts missing their beginning or end, and compared it with standard CTC by fine-tuning wav2vec 2.0 on TIMIT with partially masked labels

**Influence Network Generation for Marketing** Spring 2024
*R&D Project, Prof. Swaprava Nath, IIT Bombay*

- Generated a directed social network with influence probabilities from a banking dataset, then applied influence maximization to identify the optimal clients for advertising a new product

**EEG-Controlled Gameplay** March 2021 – July 2021
*Institute Technical Summer Project, IIT Bombay*

- Trained a CNN on EEG signals to control Atari Breakout with 75% accuracy (top 6 of 50 teams)

## Teaching & Mentoring

- Department Academic Mentor, IIT Bombay (2022–2023) — *mentored 6 sophomores*
- MA 109 Calculus I, Autumn 2021 — *Teaching Assistant, tutorials for a batch of 40 students*
- ME 119 Engineering Drawing, Spring 2022 — *Teaching Assistant, evaluated labs for 200+ students*

## Technical Skills

**Languages:** C++, Python, Bash
**ML:** PyTorch, NumPy, SciPy, Pandas
**Tools:** Git, LaTeX, ROS
**Speech:** Hugging Face Transformers, NeMo, torchaudio

## Extracurriculars

- **Topped the East Zone** in the National Financial Literacy Assessment Test (NCFE, RBI, SEBI)
- Completed one year of **Dramatics** training under NSO, IIT Bombay